

MAXIME ROCHKOULETS

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EDUCATION

KU Leuven • Leuven, Belgium

Master of Engineering • Computer Science

Thesis: "Improving Tensor Decoupling with Scaling and Smoothing"

September 2024 ~ June 2026

Provisional Grade: First-Class

University of Bordeaux • Bordeaux, France

Bachelor of Science • Computer Science

BSc obtained with honors, completed a one year exchange in Dublin during my final year.

September 2021 – May 2024

Rank: Top 10% | Grade: First Class

High School • France

French General Baccalaureate • Mathematics, English, History & Political Sciences

September 2018 – July 2021

Grade: Honors

EXPERIENCE

Genome Institute of Singapore – Researcher

Remote | October 2025 – Current

- Working part-time with the AI in Genomics research group alongside my Master's studies. Published a paper criticizing foundation models in genomics (GFM) to LMRL Workshop at ICLR 2026. [Link.] Sent a follow up of this work to NeurIPS 2026 main track.

Genome Institute of Singapore – Research Intern

Singapore | July 2025 – September 2025

- Developed fast and accurate deep learning models for real-time nanopore signals segmentation. Learned how to use high-performance GPU clusters (A100s, H100s). Was able to greatly speed up training time by multiprocessing the data pipeline. **Outperformed the existing model using a fully convolutional architecture that was more accurate and fast for both training and inference.** Learned how to write basic CUDA kernels and low level Pytorch. Finally, I presented a poster about my work at the GIS25 conference. [Link to the poster.]

Bordeaux Computer Science Laboratory – Research Intern

Bordeaux, France | May 2024 – July 2024

- My work consisted in exploring ways to train and use LLMs more effectively for software engineering tasks using efficient source code tokenizers. Learned how to build and train transformers from scratch, as well as how to fine-tune pre-trained models using PyTorch and HuggingFace. **Showed that transformer models trained using my custom tokenizers could learn classification tasks on source code while using 5 times less tokens than GPT3.** Presented my progress in weekly meetings and gave a talk in front of the entire research team at the end of my internship.

PROJECTS

fastlob

(See github.com/mrochk for more.)

- Minimalist limit order book (LOB) package in pure Python. This package is meant to be safe, clean, and easy to reuse and extend for other developers. It is a roughly 1.5k lines project that I am still developing on my free time. My goal with this package is that it one day becomes the reference for anyone who needs a reliable LOB implementation in Python. The package is published and available on PyPI.

Untangle

- A library for fast tensor decoupling algorithms in Jax. My goal is to provide the first stable and efficient implementation of several tensor decoupling methods, making it a reliable reference in the field (previously dominated by Matlab).

Quant Finance Project

- Head of Quant Research for the Beta Sigma Club in Leuven, a club where we explore topics related to Mathematics and Finance (betasigmaclub.com). This year I lead and organize our group project in which we participate to a machine learning challenge organized by the hedge fund CFM, with the goal of ranking as high as possible (currently 4th / 196).

TECHNICAL SKILLS

- Programming Languages:** Python, C, C++, Go, Java.
- Libraries:** PyTorch, Jax, Tensorflow, Numpy, Pandas, Polars, Scikit-Learn, Spark.
- Others:** Linux (personal OS), Git, Bash, gRPC, REST, Google Cloud, AWS, Docker.

LANGUAGES

- Spoken Languages: **English** [IELTS Academic: 7.5], **French** [Native] and **Russian**.

EXTRA CURRICULAR ACTIVITIES

- Football:** Played soccer with various teams for more than 10 years, competed in many competitions and tournaments in France, Ireland, Belgium and Singapore. Currently playing for KU Leuven's university team.
- Student Jobs:** Worked during every holidays since my first year of high school, mostly in restaurants and bars.
- Volunteer Work:** After school, I used to help primary school children with their homework and exercises.
- Interests:** Artificial Intelligence, French & Russian Literature, Philosophy, Finance, Blockchain, Chess, Hiking.